Notes from Ben, not part of the talk

- Side chain: now explained on slide 2 in one clause (Ben, 14 October). The whole deck was revised on 14 October along the four facts of slide 6, see `WORK-SPLIT.md`.

Highly accurate protein structure prediction with AlphaFold

Paper by Jumper et al., *Nature* 596 (2021)

Evita Bosch Pokunevich Ben Bullinger

Machine Learning for Genomics (263-5351-00L), ETH Zürich, Fall 2026

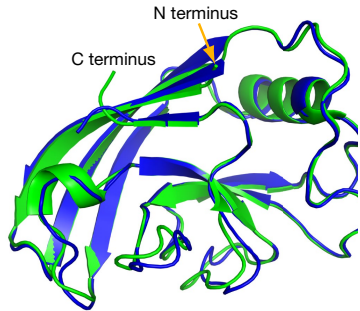
Outline

1. The problem
2. The benchmark
3. The network
   - The data
   - The network, in detail
4. Training
5. What makes it work
6. Conclusion
7. Questions

Sequence in, structure out

SIFSYITESTGTPS…

sequence of amino acids. Each has the same backbone and its own side chain



3D coordinates of all heavy atoms
Prediction in blue, experiment in green

An experiment takes months to years

Billions of sequences are known

The problem is more than 50 years old

Sequence and structure are the CASP14 target T1049, PDB 6Y4F. The structure is Fig. 1b.

The Benchmark: CASP14

CASP14: Critical Assessment of protein Structure Prediction

May-July 2020

Blind test - experimental structures are not revealed until the end of experiment.

Å (angstrom): a unit of length equal to 10^{-10} meters.

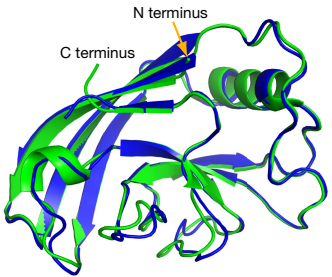
Smaller RMSD means a closer structural match.

Show the average prediction hist from the paper

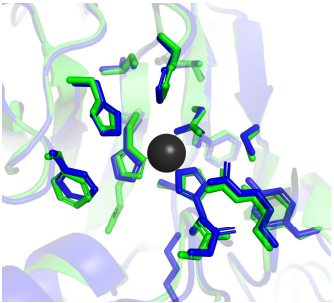
The median backbone error is below one atom width

Redrawn from Fig. 1a. The error is the Cα r.m.s.d.$_{95}$ on 87 domains, with 95% confidence intervals.

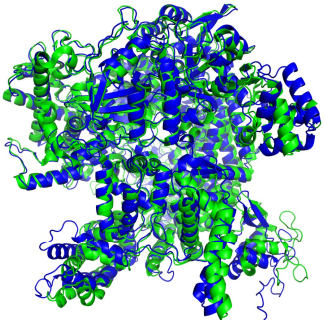
Predictions on top of experiment – delete?



Backbone



Side chains

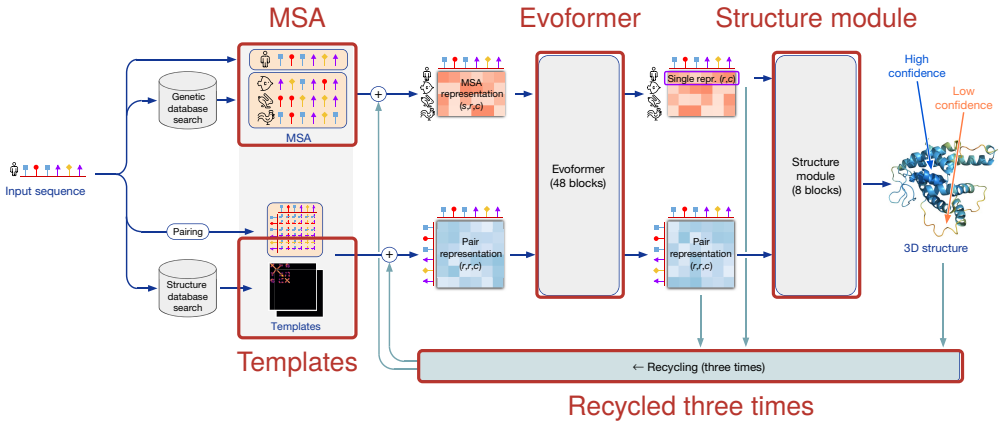


A 2,180-residue chain

AlphaFold experiment

Prediction and experiment overlap, also in the side chains

One pipeline, end to end



Coordinates come out of the network itself

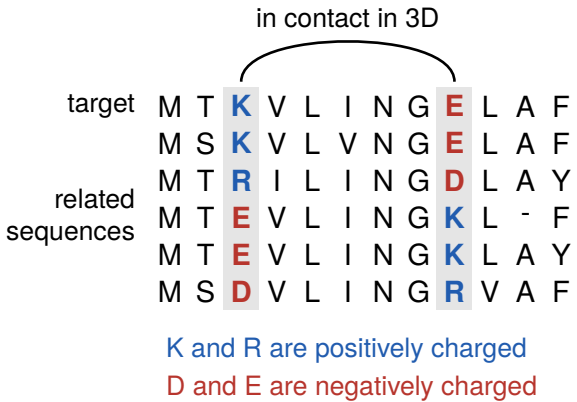
Four facts about proteins, four design choices

		Fact	Design choice in AlphaFold 2
1		An MSA is a set of sequences. A protein is a chain of residues.	Attention over the sequences themselves, along the rows and the columns.
2		Distances in 3D must close triangles.	The pair representation is updated through triangles.
3		A structure is the same after any rotation or translation.	One frame per residue. Attention uses only distances in the frames.
4		A protein and its mirror image are different molecules.	The loss compares atoms in local frames. A mirror image is penalised.

Removing any one of the four costs accuracy. The ablations measure how much.

The input

The input: MSA representation



A row is a related protein

A column is an aligned position

Contacts may change together

The network reads the sequences themselves

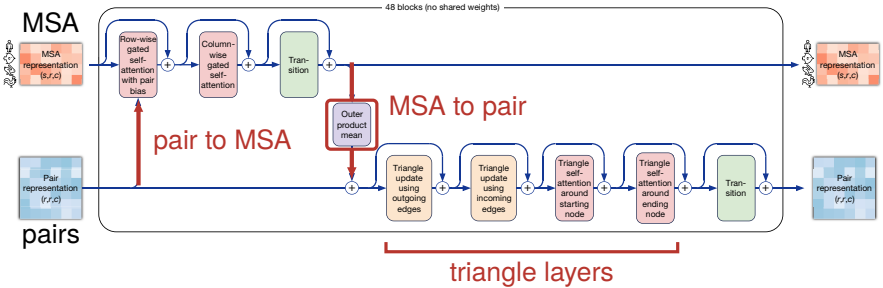
D. S. Marks et al., *PLoS ONE* 6, 2011. The alignment is an illustration, not data.

The input: Pair representation

Evoformer block



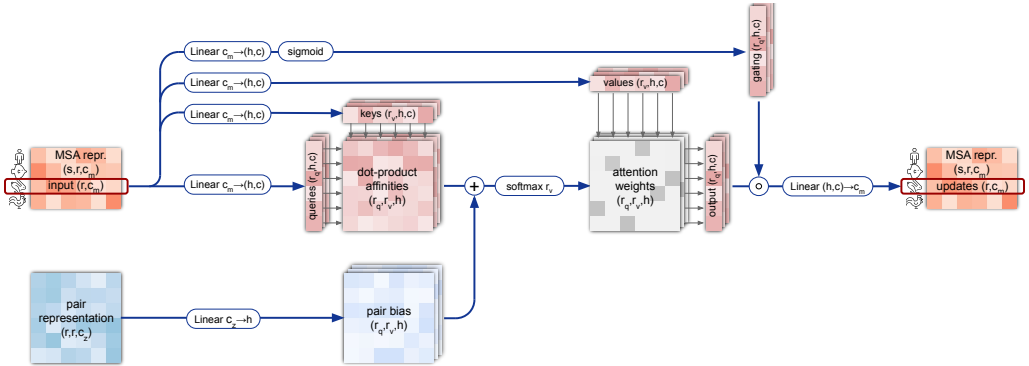
1. The sequences in an MSA are a set. The residues are a chain.



Attention along the rows and the columns. The 48 blocks do not share weights

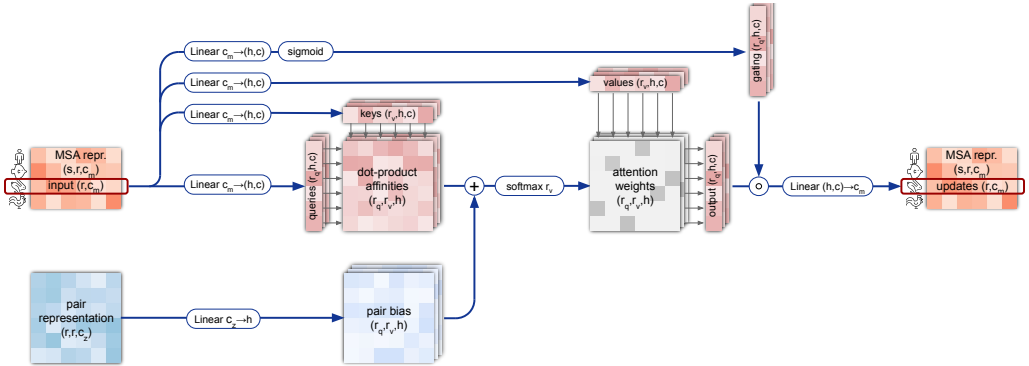
MSA and pair representation exchange information

Evoformer: MSA representation, row col



Coordinates come out of the network itself

Evoformer: MSA representation, trans out

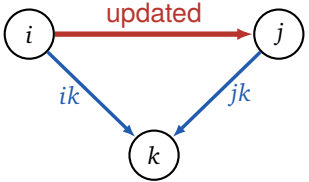


Coordinates come out of the network itself

Evoformer: Pair representation

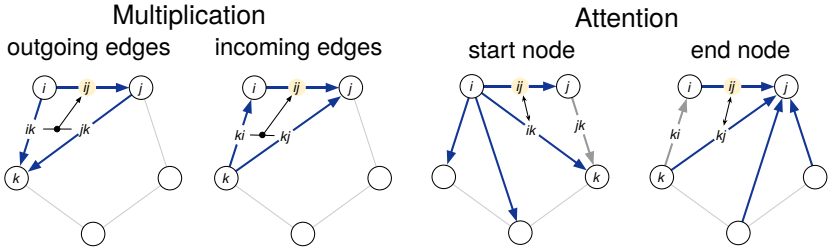
2. Distances in 3D must close triangles.





Edge ij is updated from the edges ik and jk

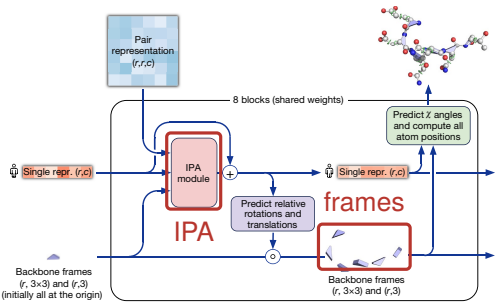
The triangle inequality acts as an inductive bias

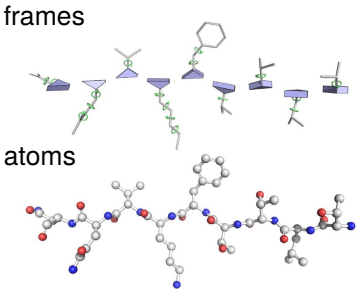


Structure module: residues as rigid bodies



3. A structure is the same after any rotation or translation.





A frame is one rotation and one translation per residue

All frames are updated at once

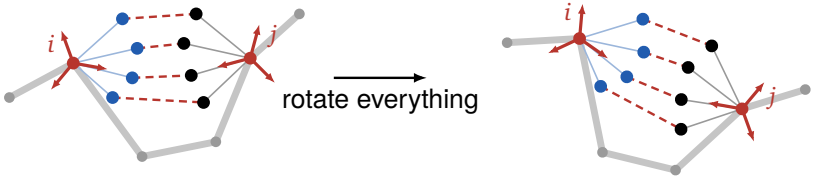
Nothing forces neighbouring residues to stay connected

Invariant point attention



3. A structure is the same after any rotation or translation.

$$a_{ij} = \mathrm{softmax}_j\Big(\underbrace{\boldsymbol{q}_i^\top \boldsymbol{k}_j}_{\text{usual attention}} + \underbrace{b_{ij}}_{\text{pair bias}} - \underbrace{\textstyle\sum_p \left\| T_i \circ \vec{\boldsymbol{q}}_{ip} - T_j \circ \vec{\boldsymbol{k}}_{jp} \right\|^2}_{\text{distances between 3D points}} \Big)$$



Points are placed in the frames of i and j

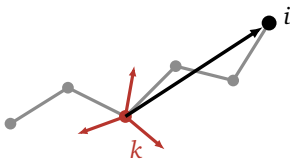
The distances stay the same

The weights are invariant to global rotation and translation

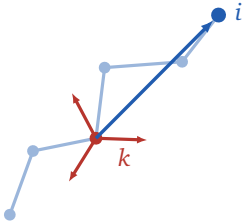
The formula is Algorithm 22 of the supplement for one head, without scaling constants.

FAPE: a loss over local frames

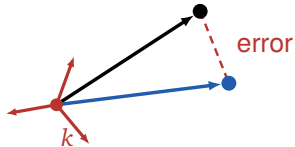
4. A protein and its mirror image are different molecules.



experiment



prediction



both, seen from frame k

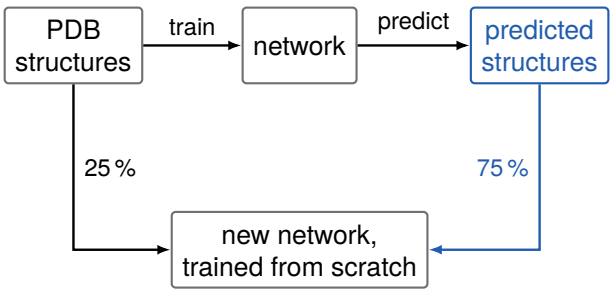
$$\text{FAPE} = \frac{1}{Z} \underset{k,i}{\text{mean}} \min\left(d_{\text{clamp}}, \left\|(\boldsymbol{R}_k, \boldsymbol{t}_k)^{-1}\boldsymbol{x}_i - (\boldsymbol{R}_k^{\text{true}}, \boldsymbol{t}_k^{\text{true}})^{-1}\boldsymbol{x}_i^{\text{true}}\right\|\right)$$

A rotation changes nothing. A mirror image is penalised.

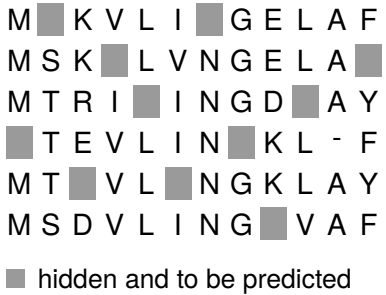
Supplementary Information, Algorithm 28, with $Z = d_{\text{clamp}} = 10\,\text{Å}$. The drawing is an illustration.

Trained beyond the PDB

Self-distillation adds 350,000 structures



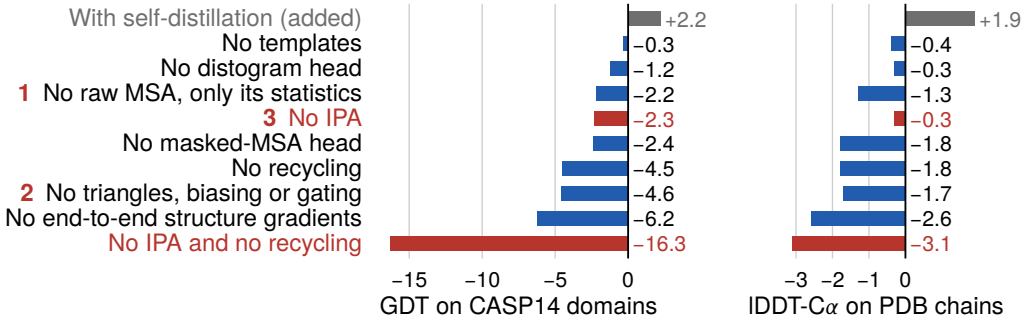
A masked-MSA loss is added



Training uses sequences without structures

Self-distillation follows Q. Xie et al., 2020. Masking follows BERT, J. Devlin et al., 2019.

Which pieces earn their accuracy

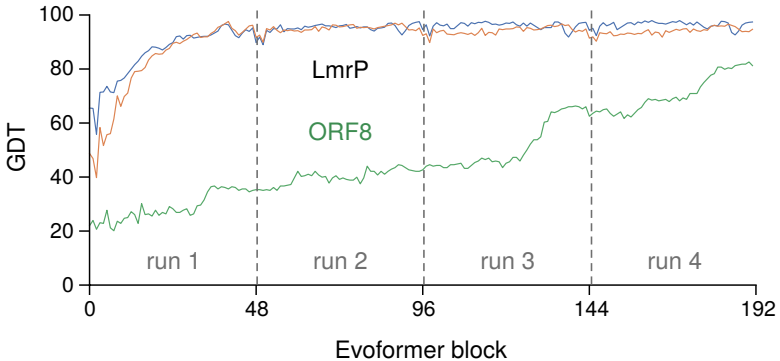


4 With a mirror-blind loss instead of FAPE, the model often builds the mirror image

Each of the four choices costs accuracy on CASP14 when it is removed

Redrawn from Fig. 4a, the mean change against the baseline over the training runs of a row. Line 4: Suppl. Fig. 9.

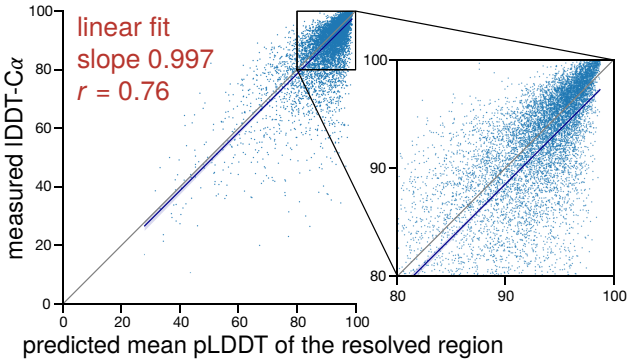
The structure forms inside the trunk



The trunk is frozen and every block gets a structure module

Some proteins are solved early and others late

The model knows when it is right

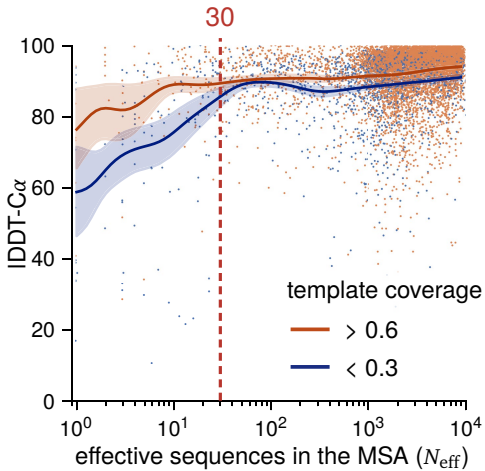


pLDDT is the predicted lDDT-Cα per residue

Predictions need a well-calibrated confidence

K. Tunyasuvunakool et al., *Nature* 596, 2021.

Where it still fails



Accuracy falls below about 30 effective sequences

AlphaFold is weaker for chains shaped by other chains

AlphaFold needs enough related sequences

What followed

2021 — RoseTTAFold appears online the same day

98.5% of human proteins are predicted

AlphaFold-Multimer predicts complexes

2022 — ESMFold uses a language model and no MSA

2024 — AlphaFold 3 is one model for proteins, DNA, RNA and ligands

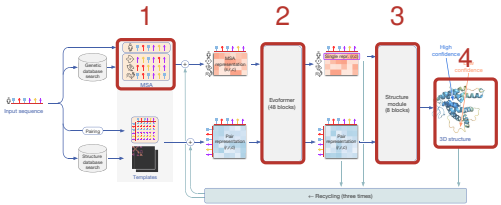
address the limits

Baker, Hassabis and Jumper receive the Nobel Prize in Chemistry

AlphaFold 3 is more accurate than specialised tools, with one exception

The first authors are Baek 2021, Tunyasuvunakool 2021, Evans 2021, Lin 2022 and Abramson 2024.

Summary



		GDT lost
1	attention over sequences	−2.2
2	triangles on the pairs	−4.6
3	frames and IPA	−2.3
4	FAPE in local frames	mirrored

Evolution and geometry are trained end to end, in about one GPU minute per model

The median backbone error is below one atom width

J. Jumper et al., “Highly accurate protein structure prediction with AlphaFold,” *Nature* 596, 583–589 (2021). GDT lost on CASP14 when removed, Fig. 4a and Suppl. Fig. 9.

Exam question

Placeholder. The question from the course team arrives about seven days before the talk.

Our answer.

Questions for the audience

Placeholder. At least two questions, then a moderated discussion.

1. First question.
2. Second question.

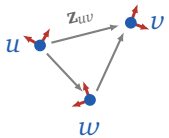
Backup: the whole network as residual updates

	Update	Message	SI
Evoformer, 48 blocks			
1	$\mathbf{m} \leftarrow \mathbf{m} + \text{RowAttention}(\mathbf{m} \mid \mathbf{z})$	node → node, edge biases the weight	Alg. 7
2	$\mathbf{m} \leftarrow \mathbf{m} + \text{ColumnAttention}(\mathbf{m})$	across the sequences at one node	Alg. 8
3	$\mathbf{m} \leftarrow \mathbf{m} + \text{MLP}(\mathbf{m})$	per node	Alg. 9
4	$\mathbf{z} \leftarrow \mathbf{z} + \text{OuterProductMean}(\mathbf{m})$	node pair → edge	Alg. 10
5	$\mathbf{z} \leftarrow \mathbf{z} + \text{TriangleMultiplication}(\mathbf{z})$, out and in	edge pair → edge	Alg. 11, 12
6	$\mathbf{z} \leftarrow \mathbf{z} + \text{TriangleAttention}(\mathbf{z})$, start and end node	edge pair → edge	Alg. 13, 14
7	$\mathbf{z} \leftarrow \mathbf{z} + \text{MLP}(\mathbf{z})$	per edge	Alg. 15
Structure module, 8 blocks with shared weights. At the start $\mathbf{s}_u = \mathbf{m}_{1u}$ and T_u is the identity.			
8	$\mathbf{s} \leftarrow \mathbf{s} + \text{IPA}(\mathbf{s}, \mathbf{z}, T)$	node → node, values moved by $T_u^{-1} \circ T_v$	Alg. 22
9	$\mathbf{s} \leftarrow \mathbf{s} + \text{MLP}(\mathbf{s})$	per node	Alg. 20
10	$T_u \leftarrow T_u \circ \text{BackboneUpdate}(\mathbf{s}_u)$	per node, in its own frame	Alg. 23
Recycling, 4 cycles			
11	$\mathbf{z} \leftarrow \mathbf{z} + \text{Embed}(d_{uv})$, C$\beta$ distances from the frames	frames → edges	Alg. 32

Every line is $\mathbf{h}_u = \phi(\mathbf{x}_u, \bigoplus_v \psi)$ on nodes or on edges. Information moves in lines 1, 4, 5, 6, 8 and 11.

Backup: message passing on a geometric graph

$$\mathbf{h}_u = \phi\Big(\mathbf{x}_u, \bigoplus_{v \in \mathcal{N}_u} \psi(\mathbf{x}_u, \mathbf{x}_v, \mathbf{e}_{uv})\Big)$$



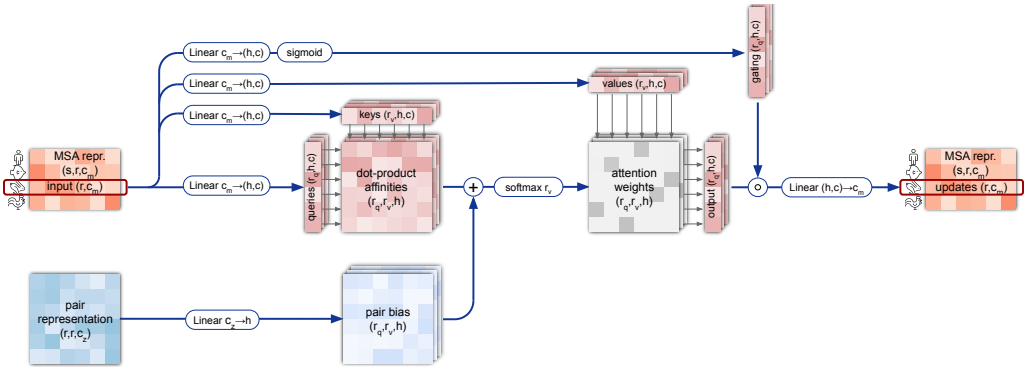
message ψ, aggregation $\bigoplus$, update ϕ (Bronstein et al. 2021).
Nodes: residues, $\mathbf{s}_u$ and frame T_u. Edges: $\mathbf{z}_{uv}$.

Block	message ψ	into
MSA row attention	$\psi(\mathbf{m}_{su}, \mathbf{m}_{sv}, \mathbf{z}_{uv})$, the edge biases the weight	node u
outer product mean	$\psi(\mathbf{m}_{\cdot u}, \mathbf{m}_{\cdot v})$, averaged over the sequences	edge uv
triangle updates	$\psi(\mathbf{z}_{uw}, \mathbf{z}_{wv})$ over every third node w	edge uv
IPA	$\psi(\mathbf{s}_u, \mathbf{s}_v, \mathbf{z}_{uv}, \lVert T_u \circ \vec{\mathbf{q}}_u - T_v \circ \vec{\mathbf{k}}_v \rVert)$, values moved by $T_u^{-1} \circ T_v$	node u
backbone update	none, ϕ alone	frame T_u

Triangles are two-hop messages. They make the edges consistent with one structure.

Geometry enters only through distances. A frame is a gauge fixed by its atoms.

Backup: MSA row attention with pair bias



Supplementary Fig. 2. Every sequence of the MSA attends along itself, the pair representation biases the weights